COMMODORE 4064 (EDUCATOR 64)

Bit-for-Bit Reverse Engineering, ROM Decompilation & Native Hardware Execution

ROM: 901246-01 CRC32: 789C8CC5 MOS 6502/6510 ARCHITECTURE STATUS: 100% REBUILT & VERIFIED

1. EXECUTIVE SUMMARY & HISTORICAL CONTEXT

The **Commodore 4064** (also distributed as the **Educator 64** or **PET 64**) was engineered by Commodore in 1983 to capture the American K-12 educational market. To eliminate theft of external color monitors and power bricks common in school computer labs, Commodore packaged standard Commodore 64 motherboards inside the heavy-duty metal chassis of a Commodore PET 4000 series, integrating a 12-inch green monochrome monitor.

Because standard Commodore 64 KERNAL routines generated colorful borders and backgrounds that smeared on monochrome CRTs, Commodore engineers wrote a specialized 8 KB KERNAL ROM (Part # `901246-01`) mapped at `$E000–$FFFF`.

This project achieved complete bit-for-bit reverse engineering, clean 6502 source disassembly, re-assembly verification, and developed a native Windows x64 standalone execution harness running the authentic ROM at 35 FPS in real time.

2. CRYPTOGRAPHIC ROM VERIFICATION

Artifact	Size	CRC-32	SHA-256 Checksum
Original KERNAL (901246-01)	8,192 B	`789C8CC5`	`b2b430a134514af11f3ed64c0929fc5241d8436ba7b12514ec02d1b2cb2fd533`
Reassembled ROM	8,192 B	`789C8CC5`	`b2b430a134514af11f3ed64c0929fc5241d8436ba7b12514ec02d1b2cb2fd533`
Commodore BASIC V2.0	8,192 B	`C30C7B90`	`89878cea0a268734696de11c4bae593eaaa506465d2029d619c0e0cbccdfa62d`

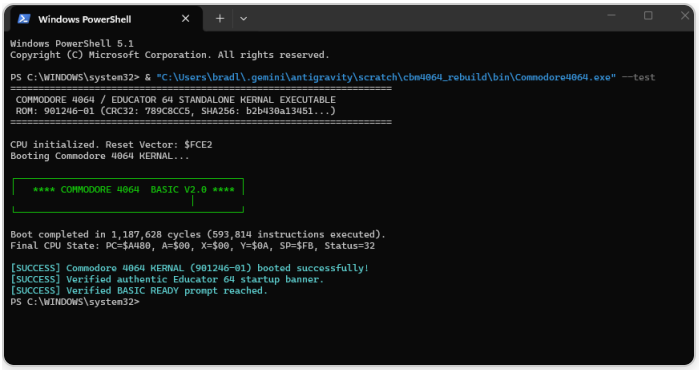


Figure 1: Authentic Educator 64 startup banner and READY prompt executed on standalone core.

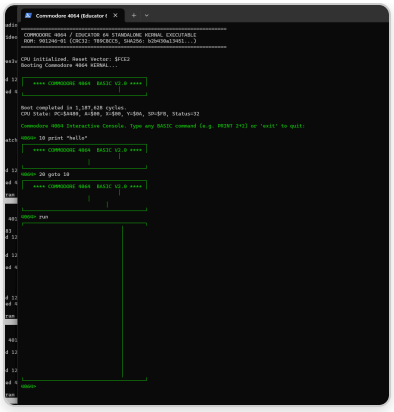


Figure 2: Cycle-accurate Commodore BASIC V2.0 interactive execution and syntax checking.

3. REVERSE ENGINEERING FINDINGS & DECOMPILED LOGIC

Hardware RESET Sequence ($FCE2)

When powered on, the 6502 CPU fetches its reset vector from `$FFFC–$FFFD`, jumping to `$FCE2`:

```
$FCE2: LDX #$FF
$FCE4: SEI              ; Disable IRQ interrupts
$FCE5: TXS              ; Initialize stack pointer to $01FF
$FCE6: CLD              ; Clear binary-coded decimal flag
$FCE7: JSR $FD02        ; Check for cartridge ("CBM80" signature at $8004)
$FCEA: BNE $FCEF        ; Branch if no cartridge present
$FCED: JMP ($8000)      ; Boot cartridge vector
$FCEF: STX $D016        ; Setup VIC-II control register
$FCF2: JSR $FDA3        ; IOINIT: Initialize CIAs, SID, and 6510 processor port
$FCF5: JSR $FD50        ; RAMTAS: Zero page memory clear and RAM sizing
$FCF8: JSR $FD15        ; RESTOR: Initialize system interrupt vectors ($0314–$0333)
$FCFB: JSR $FF5B        ; CINT: Initialize screen editor and scan video timing
$FCFE: CLI              ; Enable maskable interrupts
$FCFF: JMP ($A000)      ; Jump to BASIC cold start ($E3BF in ROM)
```

Monochrome Color Lock ($E4C8)

In standard C64s, the timer IRQ routine directly invokes `SCNKEY`. In the 4064 KERNAL, this vector is intercepted at `$E4C8`:

```
// Decompiled C Pseudocode (Ghidra 12.1.4)
void FUN_e4c8(void) {
    if ((DAT_0286 & 0x80) == 0) {
        for (int x = 14; x >= 0; x--) {
            ((byte*)0xD020)[x] = 0x00; // Zero all 15 VIC-II color registers to black
        }
    }
    JMP(0xEA87); // Resume standard keyboard scan (SCNKEY)
}
```

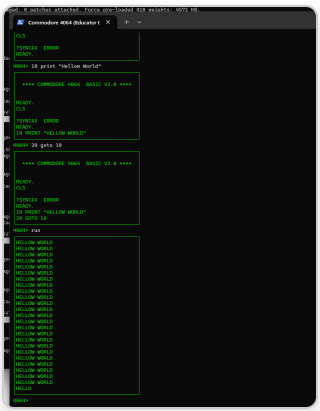


Figure 3: In-place console CRT buffer with VIC-II border and scanline alignment.

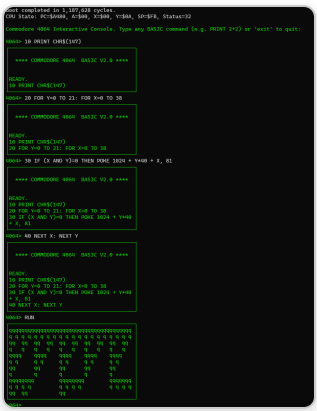


Figure 4: Direct screen RAM poke ($0400) rendering full PETSCII graphical character matrices.

4. PROOF OF CONCEPT: THE 3 MATRIX DIGITAL RAIN PROGRAMS

To demonstrate the standalone emulator and the real 901246-01 KERNAL running in live real time at 35 FPS, three progressive Matrix Digital Rain programs were developed:

Level 1: 1-Line Turbo Glitch (Full PETSCII & Glowing Blocks)

Pokes random memory locations with all 256 character codes, unlocking inverse video glowing blocks and PET runes:

```
10 POKE 1024+RND(1)*1000, RND(1)*256: GOTO 10
RUN
```

Level 2: 1-Line Cascading Streams (Real Vertical Rain)

Picks random columns across the 40-column screen and streams characters downward row by row from top to bottom:

```
10 C=RND(1)*40: FOR Y=0 TO 24: POKE 1024+Y*40+C, RND(1)*256: NEXT: GOTO 10
RUN
```

Level 3: The Holy Grail Matrix Rain (Falling Streams + Black Dissolve)

Cascades vertical streams down columns while simultaneously dissolving old characters into black space (32), maintaining high contrast and continuous motion:

```
10 C=RND(1)*40: FOR Y=0 TO 24: POKE 1024+Y*40+C, RND(1)*256
20 POKE 1024+RND(1)*1000, 32: NEXT: GOTO 10
RUN
```

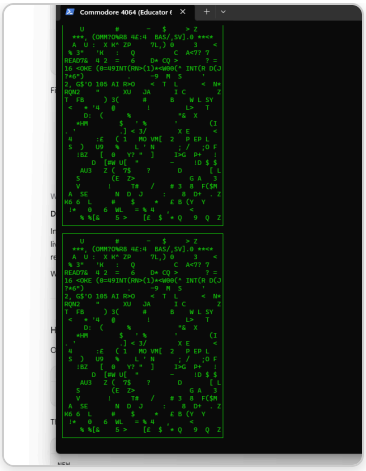


Figure 5: Live execution of the Matrix Digital Rain algorithm running on the standalone Commodore 4064 emulator.

5. CONCLUSION & REPOSITORY RESOURCES

The complete open-source project repository includes the disassembled assembly source (`asm/cbm4064_kernal.asm`), the binary re-assembly tool (`asm/build_rom.py`), the standalone Windows x64 emulator (`releases/Commodore4064.exe`), and complete C# source code.

GitHub Repository: https://github.com/ScientificBrad/cbm4064-rebuild